

EcoHawk: YOLOv11 Low-Altitude Drone System for Simultaneous Detection, Geospatial Quantification, and Risk Mapping of Invasive Species with Real-Time Authority Notification

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Abstract

In this project, the researcher developed the engineering goal of creating a real-time drone system that would fly over an area, state the location of each invasive plant, create a risk assessment map, and alert the authorities in order to reduce the number of areas with a high concentration of invasive plants. Previous studies demonstrate a step toward stronger invasive identification efforts, but all with a lack of accessibility for the average person to control, with the ability to only identify one invasive plant. The experimenter assembled a drone using a soldering iron, 4 motors, a drone frame, a custom 3D printed vibration dampening plate, and a Pixhawk and trained a machine learning model with 13,000 images to identify 16 invasive plant species. A Raspberry Pi was attached to the drone for onboard machine learning identification. An ultrasonic sensor was also integrated for obstacle avoidance. This system was tested on five different public parks and compared to expert derived ground truth data to determine how feasible this solution would be in the real world. All identifications with an interactive map are displayed on a web browser interface for authority use, with priority levels displayed for each location. From these procedures, the conclusion was developed that the drone would perform well in an actual situation, with the drone only missing 1-2 detections in each location and attaining a top-5 accuracy score of 0.89, which is higher than the average CLS model. In a real world setting, this could be used to reduce the economic burden and health impacts of invasive plant species, allowing for a transition to data driven eradication.

Introduction

Invasive species cost the world \$423 billion a year, money that could instead be allocated towards solving global healthcare issues and improving developing infrastructure (Skivington, 2025). Despite recent efforts to control the spread of invasives via citizen science apps, these plants still grow rapidly, with efforts not effective enough to halt the rise of these pests.

Despite invasives existing for centuries due to both purposeful and accidental spread, only recently has new technology emerged to help remedy the damage caused by these spreading plants. A study done by Deepti Barhate, a computer science professor at the Amity School of Engineering & Technology, demonstrated that machine learning could be used to identify plant species, especially with leaf identification to address global warming and loss of biodiversity (Barhate et al., 2024). However, this does not address the growing issue of plant recognition in the field, with many of these plants growing unchecked due to lack of knowledge on how to deploy mechanisms in this sector. In all, a system must be used that requires minimal human interference in order to alleviate the issue of lack of knowledge and inability to use existing technologies.

Literature Review

To understand the need for a drone based system that can identify invasive species from above the ground, it is imperative to understand the current literature regarding invasive species identification. An invasive plant is non-native to an area, meaning that it is not naturally grown in that location and instead was introduced. To be invasive, these plants must have harmful consequences to the environment, such as outcompeting the plants in the region. This is prevalent because invasives lack the predators local organisms have, allowing them to breed rapidly. The scale of this issue can be seen in the USDA article “Identification,” which shows

how citizen science apps like iNaturalist and Seek allow for prevention through awareness, urging citizens to use local Cooperative Extension Specialists. Citizen science apps are platforms that give ordinary people a reliable way to report invasives, allowing for contribution to research by collecting data in a safe manner (National Invasive Species Information Center, n.d.). Even though these apps have the power of rivaling social media in improving the sharing of knowledge among people, they are largely ineffective. In a study conducted by Waliczek, a Professor of Horticulture at Texas State University, when 533 Texas college students were surveyed in biology and agriculture, only 32% of the quizzes showed some level of understanding on the impacts caused by invasive species, highlighting the low level of knowledge regarding these plants (Waliczek et al., 2017). While the apps that utilize citizen help could be seen as a step in the right direction, college students were not able to identify these invasives effectively. If average citizens are relied upon for reporting, a damaged detection pipeline will result, as many of these individuals do not have the ability to differentiate among plants. This lack of public knowledge shows how the detection methods used allow for vast swaths of plants to grow unchecked, with people not being properly educated on how to determine characteristics of invasives.

With people not having the skills to accurately detect invasive plants, ground survey methods traditionally take a longer period of time to complete. However, this is detrimental to the economy, as early detection can make eradication more manageable, while late stage findings result in a higher density of invasives that must be controlled. Additionally, human surveying is slow and impractical over large areas (Perkins, 2023). Given the limitations of public reporting, solutions rooted in technology like drones are starting to become more prevalent as an alternative. In a 2023 study by Reynolds, a certified drone manufacturer, a drone was used to

identify the perennial pepperweed in California, demonstrating how cost effective drone technology would be to bring ecological safety for rapid management in wetland areas. YOLO (machine learning), a real-time object detection system with a neural network to predict the probability that an object fits into a certain class by placing a bounding box around it, was used, allowing for a targeted approach (Takekawa et al., 2023). The drone was also utilizing Pixhawk, a device that serves as the “brain” of the system, contributing flight stability. However, a large drawback of this recent technology is its inability to be used on a large scale, with the ability to pinpoint only one invasive. To be used on a practical level, drone technology must pick out species common in a particular area, as sending one drone for every species can cause damage to vegetation from the crowded technology. With the ability to detect a singular invasive, this system has not been adopted largely because it could not remedy the growing issue of multiple invasive plants populating an area. Even though research that exists justifies drone based invasive detection, a mechanism capable of multi-species simultaneous detection does not exist. This experimentation addresses the gap by developing and testing a drone system for identifying many plant species.

Research Question

Extended research must be pursued to develop a system that allows for limited human interference in invasive identification while also being practical, detecting multiple exotic species simultaneously, as the current methods available are not adequate to remedy the growing issue of dangerous plants spreading unchecked. This study will attempt to bridge this gap by utilizing a drone with real-time capabilities by asking the question: How effective is a Pixhawk system in detecting invasive plant species like the Chinese Tallow and Chinaberry from

December 2025 - February 2026 at heights below 100 feet on terrestrial grounds, as measured by the accuracy of species count compared to ground truth data?

This research is unique, as it identifies multiple invasives rather than just one while also providing immediate authority notification and risk mapping. This builds on the recent use of technology in the field of invasives, but expands this further with a more proactive approach aimed at multiple plants. One assumption of this study is that the technology will be feasible, with the Pixhawk controller providing stable flight and data capture at heights below 100 feet without interference from the wind or terrain. The study will utilize quantitative data from the YOLO classification accuracy rates and the number of correctly identified invasive plants in each testing location, which is compared directly to the ground truth (actual number) of invasives. This engineering project is vital to communities around the world, not only cutting down costs by the billions that could fund other concerns, but also by reducing habitat risks, as some invasive plants have the potential to cause wildfires. In all, the product of this experimentation can be used to further the development of proactive, data driven eradication, allowing benefits for humans, native plants, and animals.

Methodology

To test the drone system after the construction process is completed, the experimental research method is utilized. This method studies the effect of manipulating a particular variable to assess the effectiveness of a drone-based system in detecting invasive species in the real world to demonstrate the impact on detection and alert systems. This method was chosen over correlational analysis, which assumes relationships without manipulation, because experimental research allows for inferences to be made regarding the system's real effectiveness in detecting invasives. Correlational analysis can not establish a cause-effect relationship for deployment

impact. The purpose of this methodology is to create a new, drone imaging system that can detect, quantify, and create risk maps of areas with prominent invasive plant growth to reduce the damaging ecological impact these species have. Additionally, the experimental research method allows for claims to be made regarding the results of the system and supports hypothesis testing through a t-test (statistical significance).

Gather Dataset Images for Invasive Plants

Many visits are made to various parks and terrains within Texas to collect images of the following plants: Chinese Tallow Tree, Chinese Privet, Chinaberry, Paper Mulberry, Mimosa Tree, Tree of Heaven, Chinese Parasol Tree, Japanese Honeysuckle, Japanese Climbing Fern, Dodder, Kudzu, Dandelion, Dollarweed, Chamberbitter, Nandina, and Elephant Ears. By choosing plants most common in Texas, this makes the system essential to the state and provides the most targeted eradication to solve the research question. Creating the dataset also ensures lighting and angle diversity, making the CLS model more equipped to handle varying situations. Due to time constraints, the most common invasives in Texas are chosen instead of adding more plants. By using more than one species of plant, this addresses a large gap, as other similar studies with drones are only able to identify one plant, limiting practicality. A total of 13,000 images are collected to make the model more advanced, with a large quantity of each plant to allow for more ease of training.

Set Software Environment and Prepare the Dataset for Training

In Google Drive, which allows for efficient organization of each invasive into a separate subfolder, a folder called “data” is created. Then, this folder will be opened, and 16 subfolders for each of the plants are created. The YOLO model was used in a drone study in 2023, which

yielded promising results by detecting one invasive plant aerially, making it optimal to use on a larger scale (Takekawa et al., 2023).

Train and Deploy the Training Model

After all images are collected, the training program is run in Google Colab and set to 300 epochs, which is a pass through the entire dataset (see Figure 1). Google Colab runs on the Google servers, so a software powerful enough to run that code is not needed, simplifying the data collection process (Ragula, 2024). The 300 epochs are necessary to capture nuances in each invasive, as many iterations can help analyze the different lighting and angle conditions present in each invasive plant (see Appendix A).

Figure 1

Google Colab Classification Model



Drone Assembly

For constructing the drone, the drone arms and frame make up the base of the system. This structure is more lightweight because of its carbon fiber construction, allowing for stability while in flight and the capability to carry a USB camera (Lintech, 2025). Then, the frame is assembled by adding four 2212 brushless motors (two clockwise and two counterclockwise), as their 920KV rating balances the flight time and ensures the drone can maneuver in outdoor areas

effectively. Two propellers must go clockwise and the others counterclockwise to cancel out rotational torque, as per Newton's Third Law. Each motor is secured to a drone arm with M3x8 screws, and the screw holes are aligned properly to prevent vibration failures. Power distribution occurs in the Power Distribution Board (PDB) to distribute high voltage from the battery to prevent damage to secondary components. The XT60 connector is soldered onto the board after tinning the pads and wires with flux at 350°C for lead free solder to prevent overheating parts of the system. The arms of the drone are arranged in the "X" position, alternating between clockwise and counterclockwise from front to back, and the top plate is secured with four M2.5x6 mm screws and the bottom plate is attached with two screws per each arm, with these screws being increasingly tightened (see Figure 2). The Pixhawk 2.4.8 flight controller is mounted on the dampening plate using double sided tape, with the arrow on the Pixhawk pointing forward for correct orientation that allows for accurate altitude. This controller is superior in flight stability and navigation with its 32 bit processor that supports complex software, making it more ideal for mapping and photography (PX4 Development Team, n.d.). The dampening plate is created through the use of Tinkercad, and is 3D printed with the Bambu Labs P1S model (see Figure 3). The infill is set to 15% and the infill pattern is set to gyroid.

Figure 2

Drone Base Construction

**Figure 3**

Completed 3D Printed Vibration Dampening Plate



Connect and Secure Drone Electronics

For each motor, the three ESC wires that connect to the motors are connected at random, as this can only be fixed and readjusted during the calibration process, and the ESC signal wire is placed through the vibration dampening plate and plugged into the Pixhawk main output port, with the black wire facing up. Two top plate screws are removed and the GPS bracket is installed by aligning with these two missing screws. Then, it is resecured, with this new elevation in the GPS module improving satellite visibility and reducing interference from the drone frame. The GPS wires are plugged into the GPS and I2C ports of the Pixhawk, enabling position hold and compass data to be sent back to the mission planner. The receiver with the PPM reduces wire complexity and transmits to several channels while also connecting the drone to the controller. The safety switch is connected to its port and secured with three zip ties to provide a disarming device, preventing the propeller from activating accidentally and harming others (see Figure 4).

Figure 4

Drone Main Hardware and Components Assembled

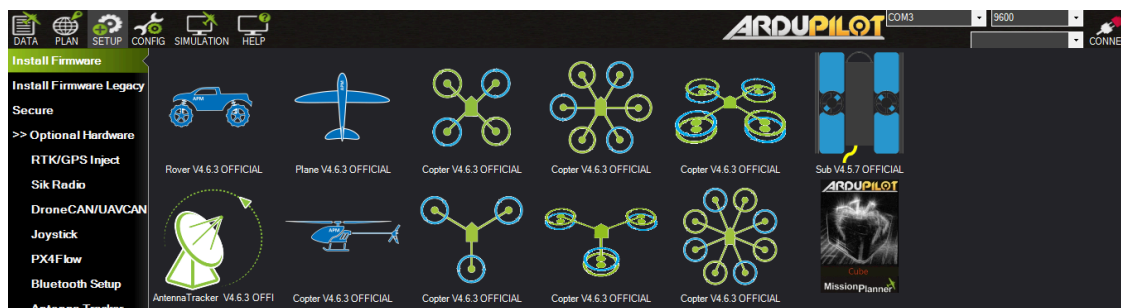


Drone Mandatory Calibration

Mission Planner is used by connecting the drone to the computer through a micro USB. This is done to conduct mandatory calibrations, such as setting the frame type to the quadcopter shaped as an “X,” which is low cost and can be moved easily, making flight with a camera more simple to operate (see Figure 5). The drone passes through slow rotations to calibrate the compass for accurate sampling of the magnetic environment in all three axes.

Figure 5

Mission Planner Calibration Phase



USB Camera, Raspberry Pi, and Ultrasonic Sensor Integration and Propellers Installation

An ultrasonic sensor is used for object avoidance to prevent drone crashes during invasive detection. Four male to female jumper wires should be plugged into GND, ECHO, TRIG, and Vcc. The wire in Vcc is connected to GPIO 5V (Pin 2) on the Pi. For TRIG, the other end of the jumper wire is connected to GPIO 23 (Pin 16). For the ECHO pin, the connection must pass through a voltage divider to step down the 5V signal. The ECHO pin is connected to a 1kΩ resistor. The other end of this resistor connects to GPIO 24 (Pin 18) and a 2kΩ resistor. The other end of the 2kΩ resistor connects to the ground line. The sensor’s GND pin is connected to the Raspberry Pi’s GND (Pin 6), making the ground line from the 2kΩ resistor tie to the same ground connection. A Python script creates the sensor object and allows for its use (see Appendix B). The USB camera is glued to the Raspberry Pi, allowing for onboard invasive

species detection by loading the trained model onto the Pi (see Appendix C). A separate battery pack gives power to the Raspberry Pi to allow for longer flight times. The USB camera, Raspberry Pi, ultrasonic sensor, and additional battery pack must be stored in a separate compartment to fix the drone's center of gravity, as a drone that is top-heavy is more likely to flip and be unstable. A container of dimensions 9in x 9in x 1in is 3D printed on a Bambu Labs P1S printer with the PLA material to allow for a stronger holding capacity for the various drone components (see Figure 6). The Pi is connected to the Pixhawk controller with a USB, giving power to the Pixhawk which is connected to the GPS module. This allows for the system to be connected, making the GPS give coordinates of an invasive plant while in flight for display on the Raspberry Pi (see Appendix D). The *detect.py* script opens the camera stream and checks for latency, with resolution lowered to minimize lag and allow for real time detection (see Figure 7). The propellers are installed by hand tightening on top of each motor, which produces a thrust force that allows the drone to lift upward.

Figure 6

Drone Construction Completed



Figure 7*USB Camera Feed****Field Testing, Mapping, and Reporting***

Trials are conducted in five different public parks, with 250 m x 250 m areas defined and checked by an environmental expert for the ground truth number of invasives, providing a benchmark to check the model's accuracy in real world conditions. This area is chosen for its large size that does not cross over significant vegetation, providing access to a variety of invasives. The drone is put into flight and the detections from the USB feed are logged with coordinates into the Pi, which provides location specifications. The path is created by going onto Mission Planner and pressing the "Plan" tab. Any points can be selected on the map, with one point set to "Takeoff," one set to "Return to Launch," and the others serving as "Waypoints." The "Data" tab is then selected, and by pressing "Arm," the drone mode is set to "Auto," allowing for the flight for each of the five park locations. Every image processed by the USB camera is coded into one of the invasive plants or as "no detection" based on the trained model. The weighted risk score is calculated by using the formula:

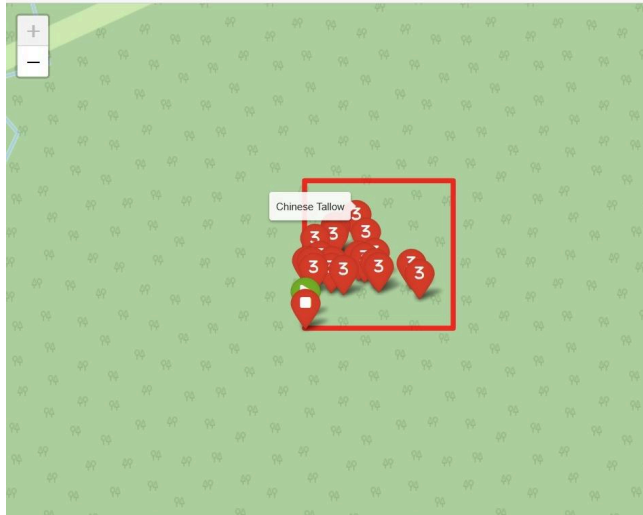
Weighted danger score = $\sum (Count_i \times Danger\ score_i)$. A danger score from 1-3 is assigned

to each invasive (3 = high impact, 2 = medium, 1 = low) to prioritize species that can result in more ecological damage (see Appendix E). If the weighted danger score is greater than or equal to 40, the area is priority risk and an alert is sent to the authorities. Each danger score and threshold is created by the field ecologist to provide credibility on immediate risk areas. If these thresholds are created arbitrarily without an advisor, an area that is high risk may be considered low risk according to the drone system, allowing invasive vegetation to spread. Maps are created by extracting data from Mission Planner and using the Python library Folium, which provides an efficient way to create geospatial visuals from raw data on a website browser, making it superior to other mapping platforms. Risk zones are marked with one of three colors: red, green, or yellow. If the location exceeds the priority risk threshold, it will be bound in red on the map (see Figure 8). Once a detection occurs, the captured image file is uploaded to Firebase Cloud Storage, while alert data including GPS coordinates, timestamp, and confidence score is logged into Cloud Firestore database. The integration powers a web dashboard that includes risk assessment maps and alerts for each drone location. When a new alert document is made in Firestore, the function triggers a detailed alert on the webpage. Other drone models do not directly alert eradication services, limiting the efficiency of these models as compared to this system, which enables instantaneous alerts. While the drone is in flight, the GPS coordinates and invasive species type are written onto a file in an SD card so when the Raspberry Pi is connected back to WiFi, the risk assessment maps can be created and the alerts can be made through Firestore. A comparison table of detected vs. ground truth counts is made for all five sites, allowing for evaluation of the model's performance through a paired sample t-test, which compares the mean of drone classifications to the ground truth mean. This gives quantitative data to determine if the results are significant or due to chance. All procedures are performed safely

and comply with ethical standards, with the expert advisor ensuring no harm is done to any plant life and no plants are uprooted or modified due to experimentation, as only aerial imagery is taken.

Figure 8

Folium Risk Assessment Maps



Results and Analysis

Field Testing with Ground Truth Comparison and Identification Accuracy Rates

Conducting field tests evaluated the performance of the drone by comparing the number the system was able to classify to the ground truth data with the number of invasives in a bounded off location. The identification accuracy was determined as the percent of ground truth plants that were correctly classified, and the weighted risk scores were found by summing the danger score of each invasive found. Alerts were sent out based on the priority threshold defined previously that makes an area high risk.

Figure 9*Ground Truth and Drone Classification Comparison Table*

Testing Location	Expert Count (Ground Truth)	Drone Classifications	Identification Accuracy (%)	Weighted Risk Score Found by Drone	Alert Sent by Drone
Location #1	22	20	90.90%	60	YES (Priority)
Location #2	18	17	94.40%	51	YES (Priority)
Location #3	15	13	86.70%	29	NO (Monitoring)
Location #4	15	14	93.30%	37	NO (Monitoring)
Location #5	24	24	100.00%	72	YES (Priority)

After all five locations were surveyed, a total of 94 invasive plants were identified from the expert advisor. The drone system classified 88 of these invasive plants accurately, meaning the accuracy rate among all the locations was 93.62%, calculated as the total number of correct drone classifications divided by the expert ground truth count. Identification accuracy ranged from 86.70% at the worst location to 100% at location five, with the drone missing a total of six invasions across every site. The accuracy of this model is higher than the previous drone systems that were investigated in the literature review, heightening accuracy rates of an aerial mechanism. The weighted danger scores ranged from 29 at location 3 to 72 at location 5, with a mean of 49.8 after every location was identified, demonstrating how the issue of invasive plants is prominent among various public locations in Texas. Locations with priority alerts typically had higher mean identification accuracy (95.1%) and mean weighted risk score (61) compared to monitoring sites (mean accuracy 90%, mean risk score 33). This shows the algorithm favored sites with stronger detection confidence and density. This was done purposely, as the alerts are sent when detection is certain and the threat is highest, reducing the number of false alarms sent to the authorities. Missed detections were found among the locations but were minimal (6.38% of instances), and happened more often when expert counts were moderate rather than high. This shows the drone had strong performance in actual conditions, with credible risk alerting based on

Classification Model Accuracy: Confusion Matrix for Identifying Invasives

Figure 10

[illegible]

A sum of 149 scenarios were tested with the different invasive plant classes, as seen when summing the numbers in each box. The rows were the actual labels of each plant, while the columns stated the predicted labels. The numbers in each box demonstrated the count assigned to a certain combination. The diagonal boxes highlighted the number of invasives that were classified accurately, and they were most prominent for every class. The Chamberbitter had the highest true positive count with 18, as the vein structure on these leaves are distinctive, allowing for the classification system to have clear features to isolate. Other plants had strong results of 10 correct predictions, including the prevalent Chinese Privet and Chinese Tallow. The Dodder and Kudzu each had 9 true positives, while the Chinaberry and Dandelion had 6. The misclassifications were few in number, with only 1-2 cells not in the diagonal entries. The model had an accuracy of approximately 92%, with the performance being especially strong in plant species with sharply diverse features. Misclassifications occurred more frequently in plants with similar traits, such as the Kudzu and Dodder vine plants. This matrix demonstrated strong class accuracy of the different invasives with few errors. The use of the confusion matrix highlights how YOLOv11 delivered reliable classification that could be used for aerial invasive plant identification. The high true positive number demonstrated strong results among the different species of plants with sharply differing attributes. Since misclassifications were low in number, this highlights how the system could be used in real world applications, as the authorities would obtain accurate information from drone identification instead of getting consistent false alarms. Some of the confusions that were found with similar vines shows a limitation, as variable conditions like lighting can cause the YOLOv11 system to misclassify an object. However, when compared with ground survey methods that achieve lower accuracy rates and take 60 minutes, this system serves as a significant upgrade, scanning over an area in just 12 minutes. These

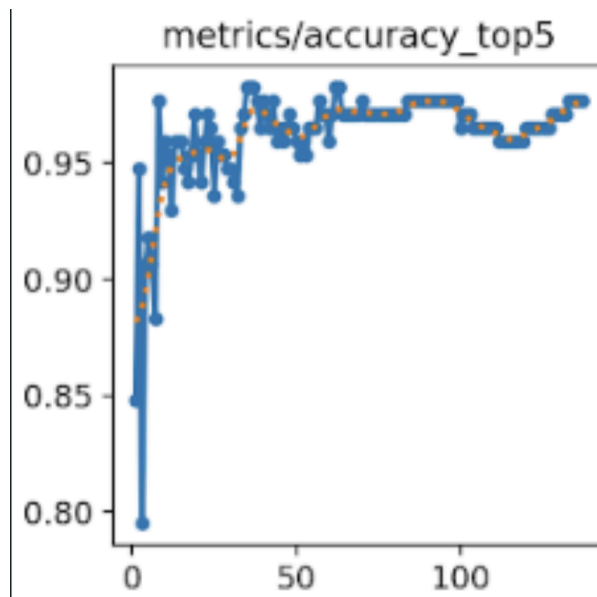
findings show the system can provide precise identification of varying plants, supporting weighted risk scores and alerting while providing support for the feasibility of the drone in real life.

Model Training Accuracy: Top-5 Accuracy Over 300 Training Epochs

One of the most essential metrics for training the machine learning model is the top-5 accuracy, which finds the percentage of cases when the correct invasive species class is in one of the model's top five predicted classes. This provides information about the machine learning model's ability to find small characteristics that define a certain plant in the dataset, especially when there is confusion with plants that look similar but the correct class is still ranked high.

Figure 11

Training Curve for top-5 Accuracy



The graph displays a blue line, representing the training accuracy values, along with orange dots, or validation top-5 accuracy at specific points. The x-axis, the number of rounds the model trains on, spans from 0 to the first 130 epochs, while the y-axis is from 0.80 to 1.00 (top-5 accuracy). When the training started, the accuracy was low (0.80) and rose significantly. By the

first 30 epochs, the training and validation accuracies are approximately 0.94, with the values that follow fluctuating but still remaining in that range. There is no large downward trend or overfitting between the training (blue line) and validation (orange dots) as the model kept training. Since the gap between training and validation averages less than 0.015, the model has looked at true features instead of memorizing noise, highlighting how efficient this would be in reality. The accuracy increased rapidly in the initial stages from 0.8 to 0.94, highlighting the effectiveness of the YOLOv11 model to train on distinct traits among the various plant species due to the strong classification framework. This marks the adaptability of this model, as if new images of various invasives were trained for a different area, 30 rounds (takes 45 minutes) can provide accuracy rates to operate the drone at a sufficient level. After rising significantly, the line plateaus with only a few deviations, maintaining a high accuracy close to 0.94. The validation points align with the training curve without any large gaps, highlighting no large overfitting (training the model on too many rounds). The accuracy remaining high after round 50 shows the ability of the system to maintain high accuracy, with fluctuations being typical in this type of training mechanism and do not indicate instability over time. The YOLOv11 model had a high top-5 accuracy during training, stabilizing at levels 0.94. Since the training and validation curves are closely aligned, this demonstrates how the parameters chosen were effective. The model can choose the correct class out of the top predictions, reducing the impact of confusions with species that look similar. This allows for real time invasive identification for a flying drone with high confidence among classes that are diverse.

Statistical comparison: Paired t-test Results

A paired t-test compared the performance of Ecohawk against the ground truth number of invasives at each location. This determined if there was a significant difference between the two samples, providing quantitative data on the reliability of the drone system.

Figure 12*Paired t-test Statistics*

p-value	0.671
t-value	0.441

The paired t-test developed a t-value of 0.441 and a p-value of 0.671, with the t-value being positive but small in number, highlighting how the mean difference between the paired observations was only slight. The p-value was higher than the 0.05 significance threshold, meaning the difference was not statistically significant. Since the p-value was above 0.05, the null hypothesis that the mean difference between the drone classification and the ground truth was statistically significant was rejected. The high p-value shows how differences align with random inconsistencies and not actual bias. The paired t-test results demonstrated that EcoHawk has similar accuracy to the ground truth itself, meaning there was no significant difference in paired comparisons. The two values show how the drone detections are reliable to expert validation, highlighting how YOLOv11 had outcomes that were statistically equal to the expert ground truth. The high field accuracy (93.62%), confusion matrix analysis (92% accuracy), training stability, and statistical testing ($p > 0.05$), demonstrate how this system would function reliably in an actual setting. This aligns with the high accuracy that can be seen in the confusion matrix, strengthening the reliability of providing accurate results in a real world scenario. For a state agency that is adopting this drone, this evidence shows how the project performs well

across all metrics instead of just one, showcasing adaptability. These statistics show that EcoHawk is credible for field deployment in the identification of various plants.

Limitations

While EcoHawk performed well in invasive species classification and mapping, there are limitations in this experimentation that should be noted. The dataset for some specific classes was not large enough, as there was confusion in plant species that looked similar to one another. This could have real world implications, as confusing various plants can manipulate the danger score of an area, which could cause the authorities to not consider an area high risk when it is in need of attention. Data collection for plants like the Kudzu and Dodder during their stages of flowering would reduce the confusion among them, as this would provide for more distinctive characteristics. This image data can be expanded easily to allow for higher accuracy. Field testing only occurred at five different locations, which may not address situations in more diverse environments and weather constraints. When deployed in another location, this could result in the system not generalizing in new ecosystems, developing lower accuracy. Expanding this testing to 15 sites, 5 sites in three different Texas regions (Piney Woods, Edwards Plateau, and Gulf Coast), would make the validity of the drone system more strong. The paired t-test showed no significant difference from the ground truth and drone classified species, but the small sample size may not be able to pick up on small biases. A t-test with the field accuracy, confusion matrix, and training metrics present how each method's limitations are combatted through the strengths in others. The drone was also sent out in more controlled scenarios, not testing actual operational situations that may introduce hardware constraints and processing issues. Future users could rely on this system, as there was limited confusion among plants and the detection

accuracy was high. However, further research with larger datasets and more trials would strengthen the strength of this system as an invasive eradication resource.

Conclusion and Future Directions

The researcher developed a real time drone system with a USB camera and YOLOv11 machine learning model that was trained on 13,000 images to identify 16 of the most prevalent invasive plant species in Texas. Previous drone systems only have the ability to detect one invasive plant, marking a significant improvement for invasive detection through efficiency, as having a drone swarm for each plant would be impractical. Field test trials were conducted in five different public parks with an identification accuracy of 93.62%, identifying 88 out of the 94 plants that were verified by a field ecologist. This allowed for the creation of reliable risk assessment maps through the use of Folium and alerts to the authorities when the danger score was larger than 40, giving priority to plant species that were more high in risk. This addresses a large need in the ecological field, as existing drones can not detect multiple invasives or alert authorities in real time. EcoHawk overcomes limitations present in citizen science apps and ground survey methods, serving as an improvement to the body of knowledge by allowing for data driven eradication that is proactive. Accurate detection through this system can reduce the cost these plants pose to the environment and protect the native vegetation in an area, allowing money to be funneled to more pressing economic sectors. While the testing process did show the feasibility of creating drone hardware cheaply that could identify invasives with high accuracy, field testing did show limitations that could be improved upon in the future. The dataset can be expanded to include images from different stages of growth from each plant, such as the flowering stage, which provides for more distinct characteristics that can improve model accuracy. This was not done initially due to time constraints of image collection. Additionally,

the use of IoT (Internet of Things) technology, a network with technology that allows for data to be exchanged across the internet, could be used to make the system more quick, especially when combined with a cellular module. By including these components, the drone would be able to identify plants more efficiently in areas with limited WiFi access, as it could use the cellular module to connect to the network and share coordinate information of each invasive through the IoT framework. Integrating this system would be expensive, exceeding the designated budget for this prototype. These components would make adoption occur faster through more efficiency, providing safety to both Texas biodiversity and human welfare.

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Appendix

Appendix A. Code Segment for YOLOv11 Model Training

```
from ultralytics import YOLO

# Load a pretrained YOLOv11 classification model (nano size)
model = YOLO("yolo11n-cls.pt") #

# Train the model on your custom dataset

# Specify the path to your dataset folder, number of epochs, and image size
results =

model.train(data="/content/drive/MyDrive/invasive_species_detection/invasive_species_images"
, epochs=300, imgsz=640) #
```

Appendix B. Code Segment for the Basic HC-SR04 Distance Sensor

```
#!/usr/bin/env python3

from gpiozero import DistanceSensor

from time import sleep

# Initialize the DistanceSensor using GPIO Zero library

sensor = DistanceSensor(echo=24, trigger=23)

try:

    while True:

        dis = sensor.distance * 100      # Convert meters to centimeters

        print('Distance: {:.2f} cm'.format(dis))

        sleep(1)

except KeyboardInterrupt:

    pass
```

Appendix C. Onboard inference loop using a custom YOLO model and camera feed

```
import cv2

from ultralytics import YOLO

import os

ROOT_PATH = '/home/yolo_env/'

BEST_PT = os.path.join(ROOT_PATH, 'invasives.pt')

model = YOLO(BEST_PT)

cap = cv2.VideoCapture(0)

while cap.isOpened():

    ret, frame1 = cap.read()

    results = model(frame1)

    for result in results:

        print(f"Top 1 class: {result.names[result.probs.top1]}, "

              f"Confidence: {result.probs.top1.conf.item():.2f}")

    cv2.imshow('YOLO V11 Detection', frame1)

    if cv2.waitKey(10) & 0xFF == ord('q'):

        break

cap.release()

cv2.destroyAllWindows()
```

Appendix D. Code Segment for Threaded Camera Stream and Pixhawk MAVLink Integration

```
import cv2

import threading

from time import sleep

import os

from ultralytics import YOLO

from pymavlink import mavutil

# === Settings ===

BORDER_SIZE = 10

BORDER_COLOR = [0, 223, 255] # Cyan border

model = YOLO('/home/sowmyavenky/yolo_env/invasives.pt')

# === Threaded Camera Class (for smoother video) ===

class VideoStream:

    # ... (initialization and background thread)

    # This reads camera frames in a separate thread so the main loop doesn't lag

# Start camera in background

stream = VideoStream(0)

sleep(1)


master = mavutil.mavlink_connection('/dev/ttyACM0', baud=115200)

master.wait_heartbeat()

print("Connected to Pixhawk!")

while True:
```

```
ret, frame = stream.read()

if not ret:

    continue

results = model(frame)

# Get prediction

confidence = results[0].probs.top1conf.item()

if confidence > 0.6:

    species_and_confidence = f"Detected {results[0].names[results[0].probs.top1]},
Confidence: {confidence:.2f}"

else:

    species_and_confidence = "

# Get data from Pixhawk

msg = master.recv_match()

if msg:

    if msg.get_type() in ['TERRAIN_REPORT', 'ATTITUDE']:

        gps_text = str(msg)


# Draw text on image

cv2.putText(frame, gps_text, (10, 30), ...)

cv2.putText(frame, species_and_confidence, (10, 50), ...)

if cv2.waitKey(1) & 0xFF == ord('q'):

    break
```

Appendix E. Weighted Danger Scores for Each Invasive

Plant	Danger Score
Chinese Tallow Tree	3
Chinese Privet	3
Chinaberry	3
Paper Mulberry	2
Mimosa Tree	2
Tree of Heaven	3
Chinese Parasol Tree	2
Japanese Honeysuckle	2
Japanese Climbing Fern	3
Dodder	3
Kudzu	3
Dandelion	1
Dollarweed	2
Chamberbitter	2
Nandina	2